

2024

MUHAMMAD TAHA

22F-3277

BCS-4C

20/04/2024

DATABASE SYSTEMS A₄

Question 1

1. Created Database with name F22_3277

2.

```
SELECT v.name, vd.contact
FROM VISITOR v
JOIN VISITOR_DETAIL vd ON v.VID = vd.VD_ID
JOIN STUDENT s ON v.STUDENT_ID = s.STUDENT_ID
WHERE s.STDNAME = 'Fatima';
```

NAME	CONTACT
Bano	0312-9876543
Bilal	0321-4567890
Bano	0345-6789123
Huma	0313-4567890
Imran	0345-6789123

3.

```
SELECT DISTINCT a.student_id AS student1, b.student_id AS student2
FROM ATTENDANCE a
JOIN ATTENDANCE b ON a.in_time = b.in_time AND a.out_time = b.out_time AND a.student_id !=
b.student_id;
```

Results Explain

no data found

4.

```
SELECT DISTINCT s.stdname
FROM STUDENT s
LEFT JOIN BILL_COLLECTION b ON s.student_id = b.std_id
WHERE b.gym_dues != 0 AND b.outstanding > b.gym_dues AND currentbillingmonth = 'March' ;
```

Results Exp

STDNAME
Fatima
Ashraf
Ayesha
Shahid
Ameer

5.

```
SELECT r.roomno, r.hostel_id, h.residency_type, h.address
```

```
FROM ROOM r
LEFT JOIN STUDENT s ON s.stdroomno = r.roomno
INNER JOIN HOSTEL h ON h.hostel_id = r.hostel_id
WHERE s.student_id IS NULL;
```

ROOMNO	HOSTEL_ID	RESIDENCY_TYPE	ADDRESS
5	2	Girls Hostel	456 Park Avenue, Lahore

6.

```
SELECT DISTINCT s.student_id, s.stdname
FROM STUDENT s
LEFT JOIN VISITOR v ON v.student_id = s.student_id
WHERE (
  SELECT COUNT(*)
  FROM VISITOR v2
  WHERE EXTRACT(MONTH FROM enter_time) = 3
  AND s.student_id = v2.student_id
) > 3;
```

Results **Expli**

no data found

7.

```
SELECT m.name, m.maintenance_date
FROM MAINTENANCE m
JOIN HOSTEL h ON h.hostel_id = m.hostel_id
WHERE h.address = '123 Main Street, Lahore' AND m.maintenance_date BETWEEN to_date('2022-02-01',
'yyyy-mm-dd') AND to_date('2022-03-01', 'yyyy-mm-dd');
```

NAME	MAINTENANCE_DATE
Cleaning supplies	02/01/2022
Window replacement	02/28/2022
Drain repair	02/10/2022

8.

```
SELECT h.hostel_ID, h.address, SUM(M.Total) AS TotalMaintenanceExpenses
FROM HOSTEL h
LEFT JOIN MAINTENANCE m ON h.hostel_id = m.hostel_id
GROUP BY h.hostel_id, h.address;
```

HOSTEL_ID	ADDRESS	TOTALMAINTENANCEEXPENSES
2	456 Park Avenue, Lahore	4200
1	123 Main Street, Lahore	4800
3	789 Broadway, Lahore	2000

9.

```
SELECT s.student_id, s.stdname, v.name AS visitor_name, v.relation
FROM STUDENT s
LEFT JOIN VISITOR v ON s.student_id = v.student_id;
```

STUDENT_ID	STDNAME	VISITOR_NAME	RELATION
S001	Ameer	Hamza	Father
S002	Fatima	Huma	Friend
S002	Fatima	Bano	Mother
S003	Ashraf	-	-
S004	Fatima	Imran	Father
S004	Fatima	Bilal	Friend
S004	Fatima	Bano	Mother
S005	Shahid	Hamza	Friend
S006	Ayesha	-	-
S007	Noman	-	-
More than 10 rows available. Increase rows selector to view more rows.			

10.

```
SELECT s.stdname, b.outstanding
FROM STUDENT s
JOIN BILL_COLLECTION b ON s.student_id = b.std_id
WHERE b.outstanding =
(SELECT MAX(outstanding)
FROM BILL_COLLECTION
);
```

STDNAME	OUTSTANDING
Ayesha	5500

11.

```
SELECT stdname
FROM STUDENT s
JOIN SHIFT_REQUESTS sr ON s.student_id = sr.requesterid;
```

STDNAME
Ameer
Fatima
Shahid

12.

```
SELECT v.name AS visitorname, v.relation, vd.contact
FROM VISITOR v
JOIN STUDENT s on v.student_id = s.student_id
JOIN VISITOR_DETAIL vd on v.vid = vd.vd_id
WHERE s.stdname = 'Ameer';
```

VISITORNAME	RELATION	CONTACT
Hamza	Father	0333-1234567

13.

```
SELECT s.stdname, s.student_id
FROM STUDENT s
JOIN GYM g ON s.student_ID = g.student_ID
WHERE g.membership_status = 1
AND NOT EXISTS
(
    SELECT 1
    FROM BILL_COLLECTION b
    WHERE b.std_id = s.student_ID AND b.currentbillingmonth = 'April'
);
```

STDNAME	STUDENT_ID
Shahid	S005
Fatima	S002
Shahid	S009
Ameer	S001

14.

```
SELECT v.name as visitorname, v.relation, v.enter_time, v.leave_time, v.leave_time - v.enter_time AS
stay_time
FROM VISITOR v
WHERE v.leave_time - v.enter_time =
(
    SELECT MAX(leave_time - enter_time)
    FROM VISITOR
);
```

VISITORNAME	RELATION	ENTER_TIME	LEAVE_TIME	STAY_TIME
Imran	Father	15-MAR-23 11.00.00.000000 AM	15-MAR-23 01.42.00.000000 PM	+000000000 02:42:00.000000

15.

```
SELECT s.student_id, s.stdname, sr.status
FROM STUDENT s
INNER JOIN SHIFT_REQUESTS sr on s.student_id = sr.requesterid;
```

STUDENT_ID	STDNAME	STATUS
S001	Ameer	1
S002	Fatima	0
S005	Shahid	1

16.

```
SELECT v1.name as visitorname, v1.relation, v1.vid AS visitor_id
FROM VISITOR v1
INNER JOIN VISITOR v2 on v1.name = v2.name and v1.relation = v2.relation and v1.vid != v2.vid;
```

VISITORNAME	RELATION	VISITOR_ID
Bano	Mother	4
Bano	Mother	2

17.

```
SELECT s1.student_ID, s1.STDName
FROM STUDENT s1, STUDENT s2
WHERE s1.stdname = s2.stdname AND s1.student_ID != s2.student_ID;
```

STUDENT_ID	STDNAME
S004	Fatima
S002	Fatima
S009	Shahid
S005	Shahid

18.

```
SELECT s.stdname, bc.currentbillingmonth, bc.total, bc.outstanding
FROM STUDENT s
LEFT JOIN BILL_COLLECTION bc on s.student_id = bc.std_id
WHERE bc.currentbillingmonth = 'March' and extract(year from bc.billpaidddate) = 2023;4
```

STDNAME	CURRENTBILLINGMONTH	TOTAL	OUTSTANDING
Ameer	March	5500	1000
Fatima	March	5500	4500
Ashraf	March	5500	3200
Fatima	March	5500	3300
Shahid	March	5500	1600
Ayesha	March	5500	5500
Noman	March	5500	0
Shahid	March	5500	300

19.

```
SELECT s.stdname
FROM STUDENT s
LEFT JOIN VISITOR v on s.student_id = v.student_id
LEFT JOIN BILL_COLLECTION b on s.student_id = b.std_id
WHERE b.paid = b.total and b.currentbillingmonth = 'March' AND v.enter_time IS NULL;
```

STDNAME
Noman

20.

```
SELECT s.student_id, s.stdname, count(v.vid) as visitorcount
FROM STUDENT s
LEFT JOIN VISITOR v on s.student_id = v.student_id
GROUP BY s.student_id, s.stdname
HAVING COUNT(v.vid) > 1;
```

STUDENT_ID	STDNAME	VISITORCOUNT
S002	Fatima	2
S004	Fatima	3

21.

```
SELECT SUM(room.rent) as total_rent_paid_by_gym_members
FROM STUDENT
INNER JOIN GYM ON student.student_id = gym.student_id
INNER JOIN ROOM ON student.stdroomno = room.roomno;
```

TOTAL_RENT_PAID_BY_GYM_MEMBERS
66000

22.

```
SELECT name, type
FROM EXPENSES
INNER JOIN HOSTEL ON expenses.hostel_ID = hostel.Hostel_ID
WHERE hostel.residency_type = 'Boys Hostel' AND EXTRACT(MONTH FROM expense_date) = 2;
```

NAME	TYPE
Water	Utility

23.

```
SELECT h.Hostel_ID, CONCAT('$', SUM(Total)) AS Total_Expenses
FROM EXPENSES e
INNER JOIN HOSTEL h ON e.hostel_id = h.hostel_id
WHERE EXTRACT(YEAR FROM expense_date) = EXTRACT(YEAR FROM CURRENT_DATE)
GROUP BY h.hostel_id;
```

Results	Exp
no data found	

Let say year is 2022 for supposing output so than

```
SELECT h.Hostel_ID, CONCAT('$', SUM(Total)) AS Total_Expenses
FROM EXPENSES e
INNER JOIN HOSTEL h ON e.hostel_id = h.hostel_id
WHERE EXTRACT(YEAR FROM expense_date) = 2022
GROUP BY h.hostel_id;
```

HOSTEL_ID	TOTAL_EXPENSES
1	\$2000
2	\$1600
3	\$1500

24.

```
SELECT DISTINCT s1.student_ID, s2.student_ID, s1.stdroomno, s2.stdroomno
FROM STUDENT s1, STUDENT s2, ROOM r1, ROOM r2
WHERE s1.stdroomno = r1.roomno
AND s2.stdroomno = r2.roomno
AND s1.student_id != s2.student_id
AND r1.floors = r2.floors
AND (r1.roomno - r2.roomno) = 1;
```

STUDENT_ID	STUDENT_ID	STDROOMNO	STDROOMNO
S005	S001	2	1
S003	S001	2	1

25.

```
SELECT h.residency_type, h.address, r.roomno, r.rent
FROM HOSTEL h
INNER JOIN ROOM r ON h.hostel_id = r.hostel_id
WHERE r.capacity >= 2;
```

RESIDENCY_TYPE	ADDRESS	ROOMNO	RENT
Boys Hostel	123 Main Street, Lahore	2	15000
Boys Hostel	123 Main Street, Lahore	6	18000
Girls Hostel	456 Park Avenue, Lahore	3	11000
Girls Hostel	456 Park Avenue, Lahore	5	20000

26.

```
SELECT h.hostel_id, h.residency_type, SUM(e.total) as total_expenses
FROM HOSTEL h
LEFT JOIN EXPENSES e on h.hostel_id = e.hostel_id
GROUP BY h.hostel_id, h.residency_type;
```

HOSTEL_ID	RESIDENCY_TYPE	TOTAL_EXPENSES
1	Boys Hostel	6000
2	Girls Hostel	5100
3	Haunted Hostel	3500

27.

```
SELECT h.residency_type
FROM HOSTEL h
INNER JOIN EXPENSES e ON h.hostel_id = e.hostel_id
WHERE EXTRACT(MONTH FROM e.expense_date) = EXTRACT(MONTH FROM CURRENT_DATE)
GROUP BY h.hostel_id, h.residency_type
ORDER BY COUNT(*) DESC
LIMIT 1;
```

RESIDENCY_TYPE
Boys Hostel

28.

```
SELECT s.student_id, g.membership_status
FROM STUDENT s
INNER JOIN GYM g on s.student_id = g.student_id
INNER JOIN ROOM r on s.stdroomno = r.roomno
WHERE g.membership_status = 1 and r.roomtype = 'Shared';
```

STUDENT_ID	MEMBERSHIP_STATUS
S002	1
S005	1

29.

```
SELECT DISTINCT s.stdname
FROM STUDENT s
INNER JOIN SHIFT_REQUESTS sr on s.student_id = sr.requesterid
INNER JOIN STUDENT b on sr.swapperid = b.student_id
LEFT JOIN BILL_COLLECTION bc on b.student_id = bc.std_id
WHERE EXTRACT(MONTH FROM bc.billpaiddate) != EXTRACT(MONTH FROM CURRENT_DATE)
```

STDNAME
Fatima
Shahid
Ameer

30.

```
SELECT BC.STD_ID, COUNT(*) AS totalbillspaid
FROM BILL_COLLECTION bc
WHERE EXTRACT(MONTH FROM bc.billpaiddate) = EXTRACT(MONTH FROM CURRENT_DATE)
GROUP BY bc.std_id
ORDER BY totalbillspaid DESC
LIMIT 1;
```

Results Explain

no data found

Question 2

1.

```
SELECT DISTINCT bc.std_id
FROM BILL_COLLECTION bc
WHERE bc.outstanding > 0
UNION
SELECT g.student_id
FROM GYM g
WHERE g.membership_status = 1;
```

STD_ID
S001
S002
S003
S004
S005
S006
S009

2.

```
SELECT std_id
FROM BILL_COLLECTION
WHERE outstanding > 0
INTERSECT
SELECT std_id
FROM BILL_COLLECTION
WHERE currentbillingmonth= TO_CHAR(SYSDATE, 'MM');
```

Results Exp

no data found

3.

```
SELECT student_id, relation
FROM VISITOR
INTERSECT
SELECT student_id, relation
FROM VISITOR
WHERE relation = 'Friend';
```

STUDENT_ID	RELATION
S002	Friend
S004	Friend
S005	Friend

4.

```
SELECT DISTINCT roomtype
FROM ROOM
UNION
SELECT DISTINCT designation
FROM ADMIN;
```

ROOMTYPE
Cordinator
Manager
Shared
Single
Supervisor

5.

```
SELECT stdname
FROM STUDENT
WHERE student_id IN
(
  SELECT student_id
  FROM VISITOR
  INTERSECT
  SELECT student_id
  FROM GYM
);
```

STDNAME
Ameer
Fatima
Shahid

Question 3

-- creating salesperson table

```
CREATE TABLE Salesperson
(
    ID INT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    Age INT,
    Salary INT
);
```

-- creating customer table

```
CREATE TABLE Customer
(
    ID INT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    City VARCHAR(255),
    Industry VARCHAR(1)
);
```

-- creating orders table

```
CREATE TABLE Orders
(
    Ord_Num INT PRIMARY KEY,
    OrderDate DATE,
    CustID INT NOT NULL,
    SalespersonID INT NOT NULL,
    Amount INT NOT NULL,
    FOREIGN KEY (CustID) REFERENCES Customer(ID),
    FOREIGN KEY (SalespersonID) REFERENCES Salesperson(ID)
);
```

-- inserting data in salesperson table

```
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (1, 'Ahmad', 61, 140000);
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (2, 'Ali', 34, 44000);
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (5, 'Noor', 34, 40000);
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (7, 'Maryam', 41, 52000);
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (8, 'Hassan', 57, 115000);
INSERT INTO Salesperson (ID, Name, Age, Salary)
VALUES (11, 'Abdullah', 38, 38000);
```

-- inserting data in customer table

```
INSERT INTO Customer (ID, Name, City, Industry)
VALUES (4, 'Jahanzaib', 'Faisalabad', 'J');
INSERT INTO Customer (ID, Name, City, Industry)
VALUES (6, 'Mubeen', 'Peshawar', 'J');
INSERT INTO Customer (ID, Name, City, Industry)
VALUES (7, 'Musa', 'Chakdara', 'B');
```

```
INSERT INTO Customer (ID, Name, City, Industry)
VALUES (9, 'Shajee', 'Chakdara', 'B');          -- changed the id to 9 otherwise it violated constraint while
adding data in orders table
```

-- inserting data in orders table

```
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (1, '08-02-1996', 4, 2, 540);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (2, '01-30-1999', 4, 8, 1800);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (3, '07-14-1995', 9, 1, 460);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (4, '01-29-1998', 7, 2, 2400);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (5, '02-03-1998', 6, 7, 600);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (6, '03-02-1998', 6, 7, 720);
INSERT INTO Orders (Ord_Num, OrderDate, CustID, SalespersonID, Amount)
VALUES (7, '05-06-1998', 9, 7, 150);
```

1.

```
SELECT DISTINCT s.name
FROM SALESPERSON s
JOIN ORDERS o ON s.id= o.salespersonid
JOIN CUSTOMER c ON o.CustID = c.ID
WHERE c.name = 'Mubeen';
```

NAME
Maryam

2.

```
SELECT name, COUNT(o.custid) AS orders_count
FROM SALESPERSON s
JOIN ORDERS o ON s.id= o.salespersonid
GROUP BY s.name
HAVING COUNT(o.custid) >= 2;
```

NAME	ORDERS_COUNT
Maryam	3
Ali	2

3.

```
INSERT INTO highAchiever (Name, Age)
SELECT Name, Age
FROM Salesperson
WHERE Salary >= 100000;
```

Results Explain Descr

2 row(s) inserted.

0.00 seconds

4.

```
SELECT Name
FROM Customer
WHERE Name NOT LIKE '%a%';
```

NAME

Mubeen

5.

```
SELECT *
FROM CUSTOMER
WHERE city =
( SELECT city
FROM CUSTOMER
GROUP BY city
HAVING COUNT(*) > 1);
```

ID	NAME	CITY	INDUSTRY
7	Musa	Chakdara	B
9	Shajee	Chakdara	B

6.

```
SELECT name
FROM SALESPERSON
WHERE ID NOT IN (
  SELECT SalespersonID
  FROM ORDERS o
  JOIN CUSTOMER c ON o.CustID = c.ID
  WHERE c.Name = 'Jahanzaib'
);
```

NAME
Ahmad
Noor
Maryam
Abdullah

7.

```
SELECT name
FROM SALESPERSON
WHERE salary >
( SELECT salary
  FROM SALESPERSON
  WHERE name = 'Abdullah');
```

NAME
Ahmad
Ali
Noor
Maryam
Hassan

8.

ID	NAME	AGE	SALARY	ID	NAME	CITY	INDUSTRY
7	Maryam	41	52000	7	Musa	Chakdara	B
11	Abdullah	38	38000	-	-	-	-
5	Noor	34	40000	-	-	-	-
8	Hassan	57	115000	-	-	-	-
1	Ahmad	61	140000	-	-	-	-
2	Ali	34	44000	-	-	-	-